

WEEKLY MINI PROJECT – 01

Title: Cybersecurity Asset Inventory System

Problem Statement

An organization maintains several IT assets such as computers, servers, routers, switches, and software applications. Managing these assets manually makes it difficult to identify the assets, track their security status, and determine which assets require immediate attention.

Develop a **Cybersecurity Asset Inventory System** that allows a security administrator to add, search, update, delete, and display information about the organization's IT assets.

The system should classify each asset based on its type and security risk level.

Input

The system should accept the following information for each asset:

- Asset ID
- Asset Name
- Asset Type
- IP Address
- Operating System
- Owner/Department
- Risk Level
- Security Status

Asset Type

The asset type should be one of:

- Workstation
- Server
- Router
- Switch
- Application

Risk Level

The risk level should be:

- Low
- Medium

- High
- Critical

Security Status

The security status should be:

- Secure
- Warning
- Vulnerable

Sample Input

Enter number of assets: 3

Asset 1

Asset ID: A101

Asset Name: HR-PC-01

Asset Type: Workstation

IP Address: 192.168.1.10

Operating System:

Department: HR

Risk Level: Medium

Security Status: Secure

Asset 2

Asset ID: A102

Asset Name: Web-Server

Asset Type: Server

IP Address: 192.168.1.20

Operating System: Ubuntu

Department: IT

Risk Level: Critical

Security Status: Vulnerable

Asset 3

Asset ID: A103

Asset Name: Core-Router

Asset Type: Router

IP Address: 192.168.1.1

Operating System: Cisco IOS

Department: Network

Risk Level: High

Security Status: Warning

Expected Output

CYBERSECURITY ASSET INVENTORY

Asset ID : A101

Asset Name : HR-PC-01

Asset Type : Workstation

IP Address : 192.168.1.10

OS : Windows 11

Department : HR

Risk Level : Medium

Status : Secure

Asset ID : A102

Asset Name : Web-Server

Asset Type : Server

IP Address : 192.168.1.20

OS : Ubuntu

Department : IT

Risk Level : Critical

Status : Vulnerable

Asset ID : A103

Asset Name : Core-Router

Asset Type : Router

IP Address : 192.168.1.1

OS : Cisco IOS

Department : Network

Risk Level : High

Status : Warning

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Total Assets : 3

Critical Assets : 1

High Risk Assets : 1

Medium Risk Assets : 1

Vulnerable Assets : 1

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Required Repository Structure

Week-01-Cybersecurity-Asset-Inventory/

```
|
|
|— src/
|   └─ asset_inventory.py
|
|
|— data/
|   └─ assets.json
|
|
|— tests/
|   └─ test_cases.md
|
|
|— screenshots/
|   └─ 01-add-asset.png
|   └─ 02-display-assets.png
|   └─ 03-search-asset.png
|   └─ 04-update-asset.png
|   └─ 05-delete-asset.png
|   └─ 06-security-summary.png
|   └─ 07-input-validation.png
|
|
└─ README.md
```